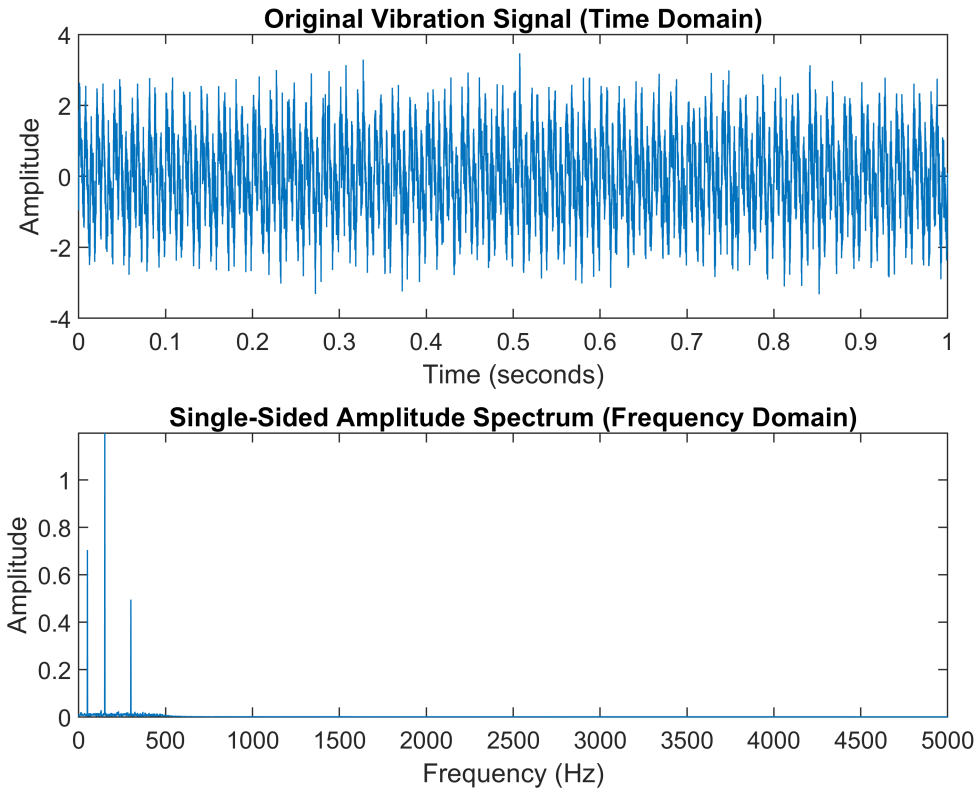
```matlab
% Vibration Analysis using DSP in MATLAB

% Step 1: Data Acquisition (Simulated vibration data)
Fs = 10000;      % Sampling frequency (Hz)
T = 1/Fs;        % Sampling period
L = 10000;       % Length of signal
t = (0:L-1)*T;   % Time vector

% Simulate a signal (sum of sinusoidal components and some noise)
f1 = 50;  % Frequency of first component (e.g., normal vibration)
f2 = 150; % Frequency of second component (e.g., imbalance)
f3 = 300; % Frequency of third component (e.g., misalignment)

signal = 0.7*sin(2*pi*f1*t) + 1.2*sin(2*pi*f2*t) + 0.5*sin(2*pi*f3*t) + 0.5*randn(size(t));

% Step 2: Signal Preprocessing - Low-pass filter to remove high-frequency noise
fc = 500;                                   % Cut-off frequency of the low-pass filter
[b,a] = butter(4, fc/(Fs/2), "low");        % 4th-order Butterworth low-pass filter
filtered_signal = filtfilt(b, a, signal);   % Apply the filter

% Step 3: Frequency Analysis using FFT
Y = fft(filtered_signal);    % Compute the FFT of the filtered signal
P2 = abs(Y/L);               % Two-sided spectrum
P1 = P2(1:L/2+1);            % Single-sided spectrum
P1(2:end-1) = 2*P1(2:end-1); % Double the amplitude except for DC and Nyquist

% Frequency vector for plotting
f = Fs*(0:(L/2))/L;

% Step 4: Plot the original signal and frequency spectrum
figure;
subplot(2,1,1);
plot(t, signal);
title("Original Vibration Signal (Time Domain)");
xlabel("Time (seconds)");
ylabel("Amplitude");

subplot(2,1,2);
plot(f, P1);
title("Single-Sided Amplitude Spectrum (Frequency Domain)");
xlabel("Frequency (Hz)");
ylabel("Amplitude");
```

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% Step 5: Anomaly Detection - Identify peaks in the frequency spectrum
threshold = 0.3; % Define a threshold for detecting anomalies
[peaks, locs] = findpeaks(P1, "MinPeakHeight", threshold);

% Print the detected frequencies
disp("Detected Frequencies (Hz) with potential anomalies:");
```

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Detected Frequencies (Hz) with potential anomalies:
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disp(f(locs));
```

```
    50   150   300
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% Step 6: Root Mean Square (RMS) Calculation
rms_value = rms(filtered_signal);
disp(["RMS Value of the Vibration Signal: ", num2str(rms_value)]);
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    "RMS Value of the Vibration Signal: "    "1.0537"
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% Step 7: Health Assessment
safe_rms_threshold = 0.8; % Example threshold for safe operation
if rms_value > safe_rms_threshold
disp("Warning: Excessive vibrations detected. Maintenance required.");
else
disp("System is operating within safe vibration levels.");
end
```

Warning: Excessive vibrations detected. Maintenance required.